Day 18 Task: VPN Fundamentals

VPN

A Virtual Private Network (VPN) is a technology that creates a protected connection between a user's device and a VPN server over a network such as the internet. It helps protect data transmitted across untrusted networks by encrypting traffic between the device and the VPN endpoint.

VPNs are commonly used for secure remote access, protecting traffic on public Wi-Fi, and connecting employees to an organization's internal network. They are an important component of network security because they help reduce the risk of unauthorized interception of data in transit.

A VPN, establishes a virtual connection through an existing physical network. It uses tunneling protocols and in many configurations, encryption to protect communication between VPN endpoints.

When a user connects to a VPN supported network traffic is routed through the VPN tunnel. Websites and other online services generally see the VPN server's public IP address rather than the user's original public IP address, although the exact behavior depends on the VPN configuration.

A VPN can improve connection privacy and security, but it does not automatically make a user completely anonymous.

How it works:

A VPN protects traffic through a sequence of steps:

- VPN connection: The user launches a VPN application and connects to a VPN server.
- Authentication: The client and VPN server verify the required credentials or certificates.
- Tunnel establishment: The VPN creates a logical tunnel using a supported VPN protocol.
- Encryption: Traffic sent through the tunnel is encrypted between the VPN endpoints when the protocol and configuration provide encryption.
- Traffic forwarding: The VPN server forwards traffic toward the destination, and responses return through the appropriate connection.

This process helps protect data from being read or modified by parties observing the network between the device and the VPN endpoint.

VPN Architecture

A typical VPN architecture consists of the following components.

- VPN Client: The application or device that initiates the VPN connection.
- VPN Server: The endpoint that authenticates clients, terminates VPN tunnels, and forwards traffic according to the configured policy.
- VPN Tunnel: The logical communication path that carries protected traffic between VPN endpoints.

- Authentication System: Verifies the identity of users or devices using passwords, certificates, multifactor authentication, or other supported methods.
- Encryption and Security Protocols: Protect communication according to the VPN protocol and its configuration.
- Destination Network: The network or internet service that the user accesses through the VPN.

Types of VPNs

Remote-Access VPN

A remote-access VPN allows an individual user to connect securely to an organization's private network from a remote location. It is commonly used by employees working from home or traveling.

Site-to-Site VPN

A site-to-site VPN securely connects two or more networks over an untrusted network. Organizations use it to connect branch offices with headquarters or data centres.

Client-to-Site VPN

A client-to-site VPN allows a device running VPN client software to connect to a private organizational network. It is a common implementation of remote-access VPN technology.

Personal or Commercial VPN

A personal VPN routes a user's traffic through a VPN provider's server. It can help protect traffic on local networks and change the public IP address visible to destination services. However, the provider may be able to observe certain connection metadata, depending on the service and configuration.

Common VPN Protocols

OpenVPN

OpenVPN is a widely used VPN protocol that supports secure tunnels using TLS-based authentication and cryptographic protection. It is flexible and can operate over UDP or TCP.

WireGuard

WireGuard is a modern VPN protocol designed to be relatively simple and efficient. It uses modern cryptographic primitives and is supported by many VPN services and network systems.

IPsec

Internet Protocol Security (IPsec) is a collection of protocols that protects IP communication. It is commonly used in site-to-site and remote-access VPNs, often together with IKE for key exchange and authentication.

IKEv2/IPsec

IKEv2/IPsec combines the Internet Key Exchange version 2 protocol with IPsec protection. It supports secure tunnel establishment and can handle network changes effectively when appropriately configured.

PPTP

Point-to-Point Tunneling Protocol (PPTP) is an older VPN protocol with known security weaknesses. It should not be used for modern security-sensitive deployments.

Security Benefits of VPNs

Encryption

VPN encryption helps protect traffic from being read by unauthorized parties while it travels between the VPN endpoints.

Secure Remote Access

VPNs allow authorized users to access private organizational resources from remote locations without exposing every internal service directly to the public internet.

Protection on Public Wi-Fi

A properly configured VPN can protect traffic from local network observers on public Wi-Fi between the device and the VPN server. It does not eliminate risks from phishing, malware, or compromised devices.

IP Address Masking

When internet traffic is routed through a VPN server, destination services generally see the VPN server's public IP address instead of the user's original public IP address.

Secure Network Connectivity

Site-to-site VPNs help organizations protect communication between separate offices and networks over untrusted infrastructure.

VPN Encryption vs. Anonymity

VPN encryption and online anonymity are different concepts.

VPN encryption protects traffic between the VPN client and the VPN endpoint, helping prevent local network observers from reading that traffic.

Anonymity refers to the ability to use a service without having one's identity or activities linked back to oneself. A VPN alone cannot guarantee this.

A VPN provider may see the user's real IP address and connection metadata. Websites may still identify users through account logins, cookies, browser fingerprinting, or other tracking methods. Additionally, DNS leaks, incorrect routing, and compromised devices can expose information if the VPN is not configured properly.

Therefore, a VPN is a privacy and security tool, not a guarantee of complete anonymity.

Limitations of VPNs

VPNs do not automatically protect users from phishing, malware, unsafe downloads, or weak passwords. They also do not guarantee that the VPN provider will never log or disclose connection information.

VPN performance can be affected by encryption overhead, server distance, network congestion, and server capacity. A VPN also protects only the traffic covered by its configuration, so split tunneling or routing errors may leave some traffic outside the tunnel.

Users should choose reputable VPN services, use strong authentication, keep VPN software updated, and verify that the configuration meets their security needs.